PATH Report

Auto-generated summary of PATH outputs

PATH is a decision-support tool that provides users with information about where to prioritize post-fire hillslope-level treatment that minimizes cost, subject to biophysical constraints. It integrates WEPPcloud's Omni Scenario Runner simulation outputs, which allows the assessment of multiple pre and post treatment scenarios. Users enter their treatment costs, landscape constraints, and biophysical thresholds and PATH identifies hillslopes for treatment using binary integer programming. This report provides a summary of PATH's outputs including tables, plots, and maps of the selected hillslopes and treatments. Figure 1 illustrates the overall workflow of PATH, from an inital WEPPcloud scenario setup and execution, to data processing and optimization in PATH, to the generation of results and this report.

Figure 1: Simple PATH workflow

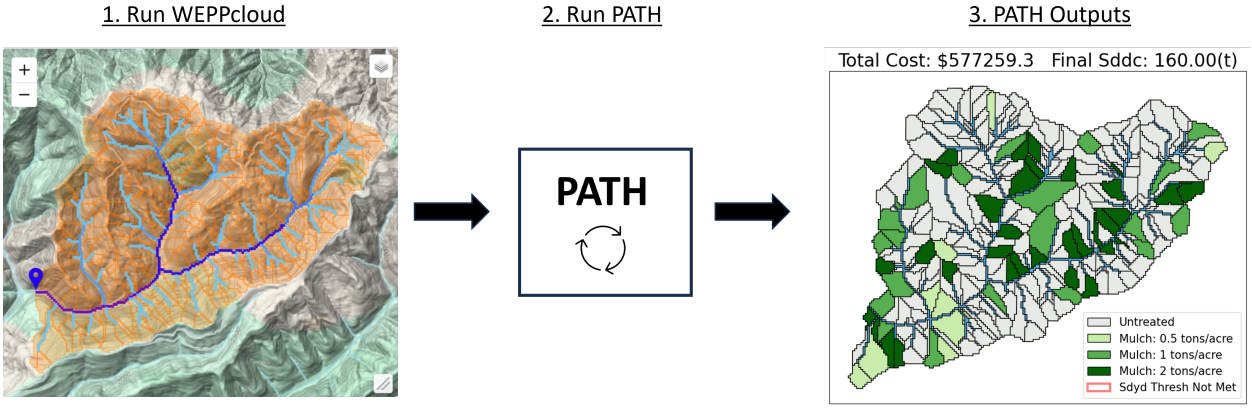


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1 Parameters

Users have the ability to input values for Hillslope Sediment Yield Threshold, Outlet Sediment Discharge Threshold, Slope Filter Range, Burn Severity filter, Treatments, and Treatment Costs. Using these parameters, along with the WEPPcloud scenario outputs, PATH identifies hillslope-level treatment assignments that minimize total cost while meeting the specified sediment thresholds at the hillslope and outlet scales. Table 1 summarizes the user-entered parameters.

Table 1: User-entered PATH parameters.

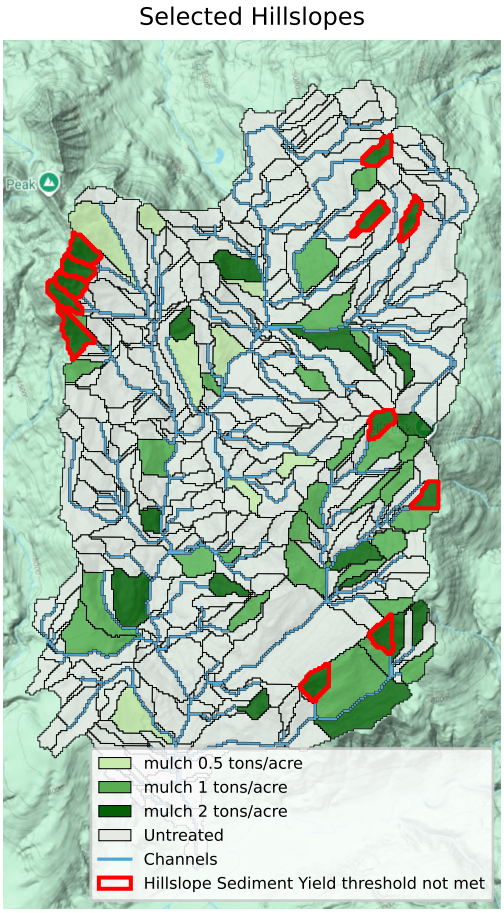
Parameter	Value
Hillslope Sediment Yield Threshold (tons/acre)	10.00
Outlet Sediment Discharge Threshold (tons)	35000.00
Slope Filter Range (degrees)	None
Burn Severity Filter	None
Treatment 1	mulch 0.5 tons/acre
Treatment 2	mulch 1 tons/acre
Treatment 3	mulch 2 tons/acre
Treatment Cost ($/acre)	1237.50, 2475.00, 4950.00

2 Run Results

2.1 Map of Selected Hillslopes

The map below shows the selected hillslopes by treatment. A hillslope outlined in red indicates that the Hillslope Sediment Yield threshold was not met even after treatment.

Figure 2: Selected hillslopes by treatment assignment. Red outlines indicate hillslopes where the sediment yield threshold could not be met.



2.2 Key Outputs

Tables 2 and 3 summarize the key outputs of the PATH optimization. Table 2 shows the total cost, total outlet sediment discharge reduction, and final outlet sediment discharge after treatment for the resulting treatment scenario. Table 3 summarizes the number of hillslopes assigned to each treatment level. To view the full hillslope-by-hillslope assignment table, please refer to the Appendix at the end of this report.

Table 2: The total cost, total outlet sediment discharge reduction, and final outlet sediment discharge after treatment, for the resulting treatment scenario.

Metric	Value
Total cost ($)	$3285338.76
Total outlet sediment discharge reduction (tons)	9007.10
Final outlet sediment discharge (tons)	34999.60

Table 3: Number of hillslopes assigned to each treatment category.

Treatment	Hillslope Count
0.5 tons/acre	8
1 tons/acre	29
2 tons/acre	26
No treatment	37

3 Sediment Threshold Analysis

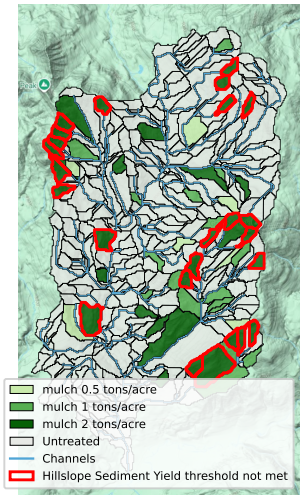
This section provides additional analysis of the Hillslope Sediment Yield and Outlet Sediment Discharge thresholds and their impact on treatment selection and costs.

3.1 Hillslope Selection Maps at Threshold Percentiles

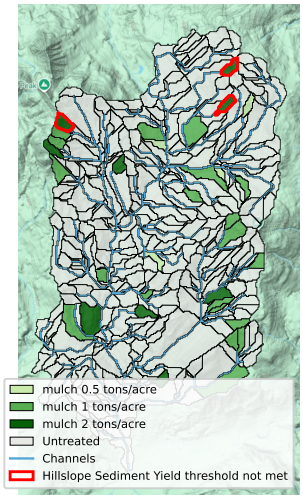
The following maps show the hillslopes selected for treatment using threshold pairs derived from the 25th, 50th, and 75th percentiles of the evaluated Hillslope Sediment Yield and Outlet Sediment Discharge threshold ranges. These maps provide a visual representation of how the spatial distribution of selected hillslopes and treatments change across the range of evaluated thresholds.

Figure 3: Selected hillslopes and treatments for the 25th, 50th, and 75th percentile sediment threshold pairs.

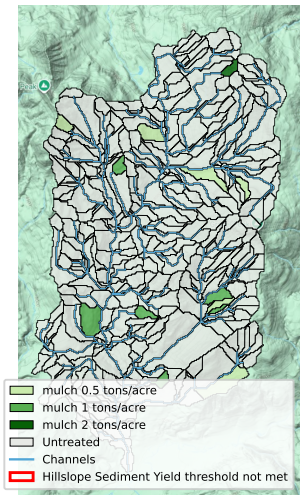
25th Percentile Threshold Pair | Hillslope Sediment Yield=7.00 tons/acre, Outlet Sediment Discharge=34926.00 tons | Cost: $3910374.52



50th Percentile Threshold Pair | Hillslope Sediment Yield=14.00 tons/acre, Outlet Sediment Discharge=37845.00 tons | Cost: $1482289.38



75th Percentile Threshold Pair | Hillslope Sediment Yield=22.00 tons/acre, Outlet Sediment Discharge=40926.00 tons | Cost: $431912.00



3.2 Cost vs. Threshold Visualizations

The following response curves show how total treatment cost changes when one threshold is varied while the other is held fixed at the user-selected value. Together they help distinguish which threshold is most strongly constraining the optimization and where additional tightening of thresholds produces disproportionately large cost increases.

The first plot isolates the effect of changing the Hillslope Sediment Yield threshold while holding the Outlet Sediment Discharge threshold constant. A steeper slope in this curve indicates that small reductions in allowable hillslope sediment yield require substantially greater investment. The second plot performs the complementary analysis for the outlet sediment threshold.

Figure 4: Total treatment cost as a function of the Hillslope Sediment Yield threshold, with the Outlet Sediment Discharge threshold fixed at the user-selected value.

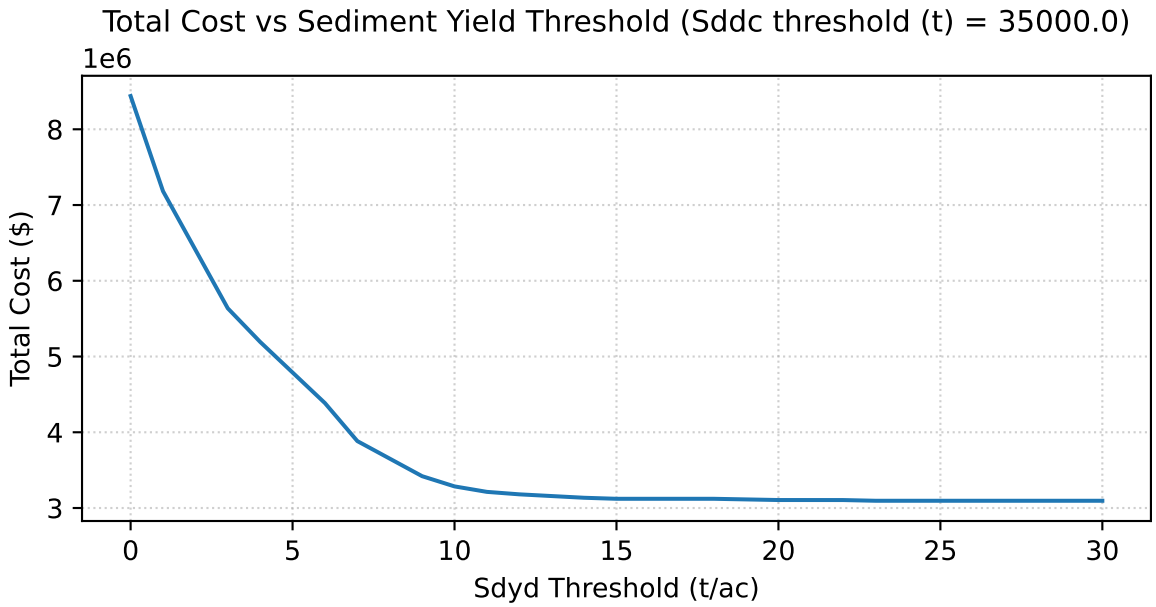
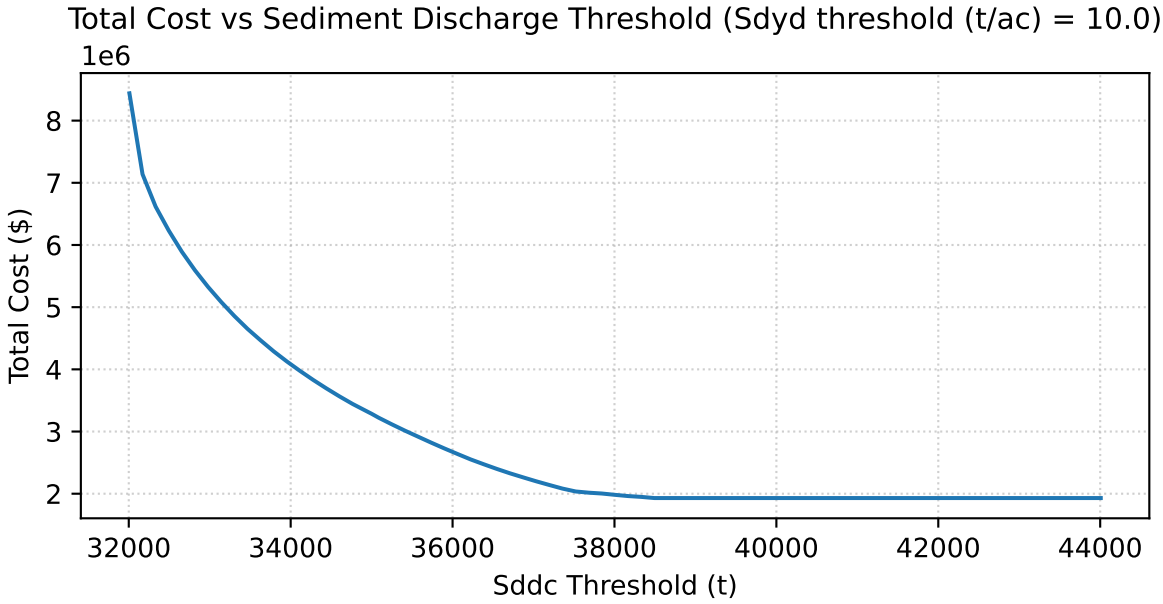


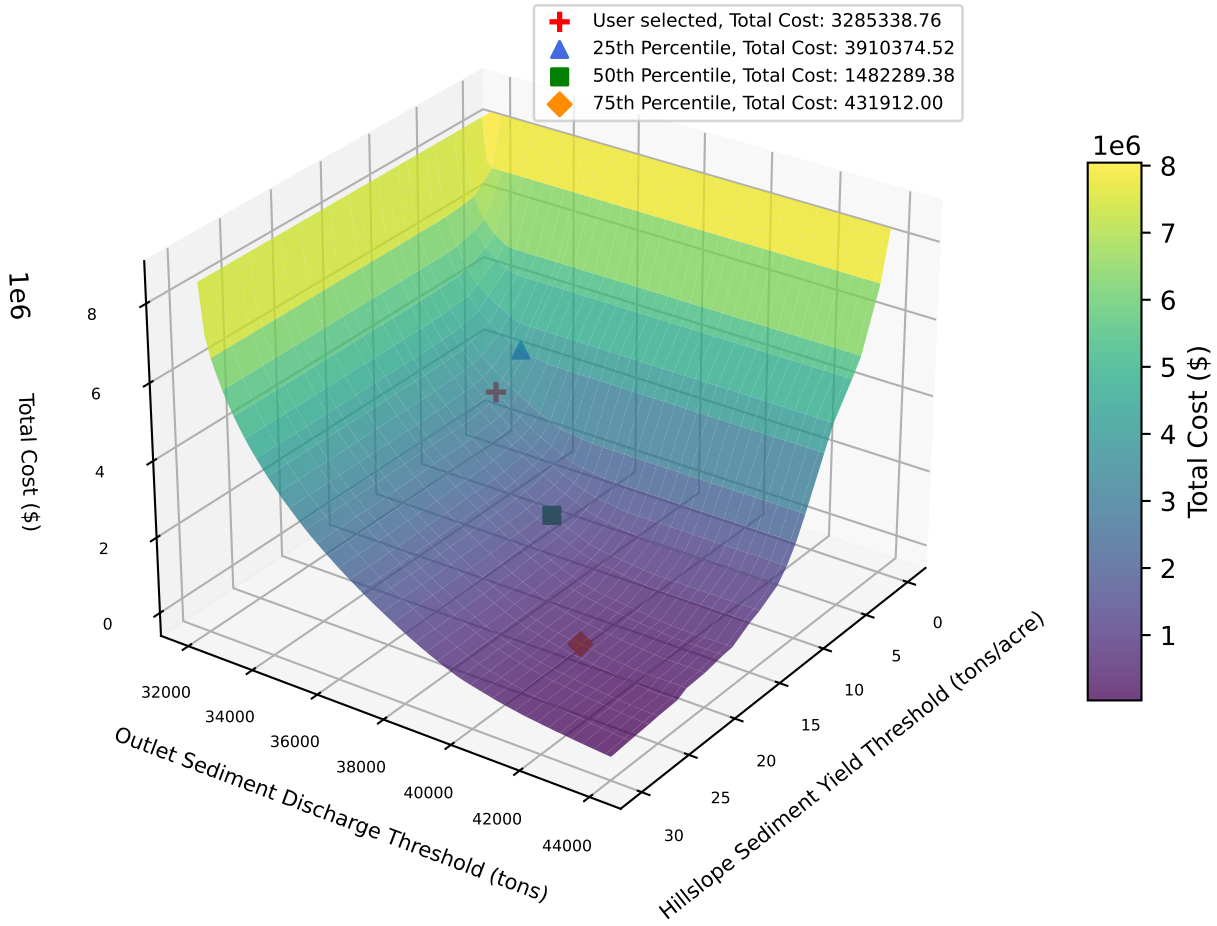
Figure 5: Total treatment cost as a function of the Outlet Sediment Discharge threshold, with the Hillslope Sediment Yield threshold fixed at the user-selected value.



3.3 Total Cost Surface

The surface below shows total treatment cost across all evaluated combinations of the two sediment thresholds. The red marker (+) indicates the user-selected thresholds. The blue (25th), green (50th), and orange (75th) markers indicate the percentile-based Sdyd and Sddc threshold pairs used in the maps above.

Figure 6: 3D cost surface with user-selected thresholds and the 25th, 50th, and 75th percentile Sdyd and Sddc threshold-pair markers.



Appendix: Hillslope-Level Treatment Results

Table 4 provides the full hillslope-by-hillslope treatment assignment and final Hillslope Sediment Yield values. Any hillslope where the final sediment yield exceeds the user-selected threshold is outlined in red. This table allows for detailed examination of how individual hillslopes were treated and their resulting sediment yields.

Table 4: Full hillslope-level treatment assignment and final Hillslope Sediment Yield. Rows outlined in red exceed the user-selected Hillslope Sediment Yield threshold.

Wepp ID	Final Sediment Yield (tons/acre)	Treatment
1	9.89	2 tons/acre
2	9.55	1 tons/acre
3	8.81	No treatment
4	5.03	2 tons/acre
5	3.33	2 tons/acre
6	0.09	2 tons/acre
7	9.19	0.5 tons/acre
8	0.83	1 tons/acre
9	2.78	2 tons/acre
10	17.52	2 tons/acre
11	11.73	2 tons/acre
12	4.71	2 tons/acre
13	6.89	No treatment
14	20.05	2 tons/acre
15	4.35	2 tons/acre
16	0.75	2 tons/acre
17	9.60	1 tons/acre
18	16.21	2 tons/acre
19	5.27	2 tons/acre
20	6.24	1 tons/acre
21	7.18	1 tons/acre
22	8.47	No treatment
23	1.56	1 tons/acre
24	5.49	1 tons/acre
25	12.68	2 tons/acre
26	11.33	2 tons/acre
27	9.77	1 tons/acre
28	11.56	2 tons/acre
29	10.44	2 tons/acre
30	0.87	1 tons/acre
31	9.18	2 tons/acre
32	10.21	2 tons/acre
33	9.39	1 tons/acre
34	4.10	2 tons/acre
35	9.48	0.5 tons/acre
36	11.04	2 tons/acre
37	10.24	2 tons/acre
38	9.95	2 tons/acre
39	10.16	2 tons/acre

Wepp ID	Final Sediment Yield (tons/acre)	Treatment
40	0.80	1 tons/acre
41	2.99	No treatment
42	9.07	2 tons/acre
43	9.35	1 tons/acre
44	8.75	1 tons/acre
45	9.19	0.5 tons/acre
46	9.97	1 tons/acre
47	7.09	1 tons/acre
48	6.54	1 tons/acre
49	2.30	1 tons/acre
50	8.22	No treatment
51	8.47	No treatment
52	6.86	0.5 tons/acre
53	8.72	1 tons/acre
54	6.81	1 tons/acre
55	6.63	1 tons/acre
56	0.87	No treatment
57	6.37	No treatment
58	4.39	1 tons/acre
59	7.80	No treatment
60	8.83	No treatment
61	0.65	1 tons/acre
62	0.48	No treatment
63	1.43	1 tons/acre
64	0.84	0.5 tons/acre
65	7.07	No treatment
66	0.97	2 tons/acre
67	6.60	No treatment
68	5.67	1 tons/acre
69	3.30	No treatment
70	2.73	No treatment
71	6.30	No treatment
72	2.66	1 tons/acre
73	5.00	No treatment
74	7.92	1 tons/acre
75	5.43	No treatment
76	2.07	1 tons/acre
77	0.77	0.5 tons/acre
78	3.87	No treatment
79	1.62	No treatment
80	5.12	No treatment
81	3.66	No treatment
82	1.09	1 tons/acre
83	0.88	0.5 tons/acre
84	1.04	No treatment
85	3.77	No treatment
86	2.79	1 tons/acre
87	0.70	No treatment
88	5.09	No treatment

Wepp ID	Final Sediment Yield (tons/acre)	Treatment
89	4.59	No treatment
90	4.28	No treatment
91	1.47	No treatment
92	1.89	No treatment
93	1.07	No treatment
94	0.65	No treatment
95	1.82	0.5 tons/acre
96	3.59	No treatment
97	5.18	No treatment
98	2.72	No treatment
99	4.39	No treatment
100	3.35	No treatment